# Saam Haghighat-Grami

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Roswell, GA
Interests – Autonomous, Mobile, Collaborative Robotics

Graduating December 2026
LinkedIn – saam-haghighat-grami
GitHub – Saam-Grami
Portfolio – saam-grami.github.io
Active Military Secret Security Clearance


## Education

- **Kennesaw State University** *January 2022 – May 2022, January 2024 – Present*
  *Mechatronics Engineering, B.S. – Graduation December 2026* CGPA: 4.00
- **Marine Corps Communication Electronics School** *August 2022 – October 2023*
  *Ground Electronics Transmission Systems Maintainer* CGPA: 4.00

## Technical Skills

**Autonomy & Robotics**: ROS2, Gazebo, Nav2, ORBSLAM3, MAVLink2, ArduPilot, Mission Planner, UAV/UGV integration, RTAB Map, UR5e, Control Theory, QGroundControl
**Embedded & Hardware**: ESP32, Arduino, Raspberry Pi 5, OpenMV, PX4, Matek H743 WLITE, LiDAR, RGB-D cameras, IMUs, ad hoc networking, Nema 23 stepper, N20 motors, MentorPi, Tello, ESP-NOW
**Languages & Tools**: C++, Python, MATLAB, LaTeX, Bash, Lua, Linux/WSL2, Git, VS Code, Arduino IDE, SolidWorks, Onshape, Docker
**Perception & ML**: PyTorch, TensorFlow, YOLO, CNNs, LSTMs, OpenCV, sensor fusion, Optical FLow
**Manipulators & Automation**: Denavit-Hartenberg parameters, forward and inverse kinematics, April tags, trajectory planning, Ladder Logic, TIA Portal, PLCs, HMI development

Portfolio — **saam-grami.github.io** — Full publication list, in-progress projects (multi-drone swarm sim, GPS-denied UAV–UGV localization), and additional technical detail.

## Research & Engineering Experience

- **MEAP – Mass Estimation Autonomous Pathfinding** *November 2025 - Present*
  *Robotics Software Architect & Project Coordinator – AgriSys Lab, Dr. Muhammad Hassan Tanveer*
  - Architected a UAV–UGV collaborative search-and-rescue system, fusing aerial map stitching with SLAM-based ground navigation to estimate obstacle mass and volume in real time.
  - Led a 5-person team building the UAV autonomy stack, validated on a 2 $m^2$ area and scaling to 25 $m^2$; leading the mass-estimation ML pipeline, targeting 85% mission success on obstacles up to 0.5 kg.
- **DARB – Drone Arm Robotic Backpack (Personal Project)** *May 2025 - Present*
  *Sole Designer and Builder*
  - Designed a backpack-mounted robotic arm that autonomously captures, stows, and relaunches a drone mid-flight, targeting a 30-second capture-to-relaunch cycle.
  - Built and flight-tested the full system with created custom airframe using ArduPilot on a Matek h743 WLITE through MAVLink. Attached OpenMV orients drone to robotic arm. Drone and Arm communicate autonomously through ESP-NOW. – across repeated live test flights.
- **Trajectory Forecasting for Floating Debris in Waterways** *June 2026 - August 2026*
  *Funded Undergraduate Researcher – Summer Undergraduate Research Program, Dr. Matt Marshall*
  - Engineered a trajectory-forecasting ensemble ML model for waterways that predicts 10 seconds of debris paths from just 1 second of video input, achieving a low average error of 75px on 1080p footage (4% of frame width).
  - Demonstrated a 55% reduction in mean forecast error over the industry-standard physics baseline (Advection-Driven Particle Transport with Inertial/Slip Correction).
- **SoilBus – Soil Monitoring Probe Network** *August 2025 - December 2025*
  *Lead Software Developer – AgriSys Lab, Dr. Muhammad Hassan Tanveer*
  - Designed an AD HOC soil-monitoring mesh network with auto-discovering probes, letting any single node offload the entire network's data.
  - Achieved full mesh connectivity in 15s and $\sim$5s per-node data offload; validated BFS-based routing on 5 physical nodes, scaling to 26 simulated.

## Publications

Author on 3 works in autonomous systems research – 1 published (IEEE SoutheastCon 2026, IEEE Xplore), 2 first-author manuscripts in preparation for peer reviewed CV venue and IROS 2027. *Full citations and DOIs at portfolio above.*

## Industry & Military Experience

- **EOSYS Industrial and Manufacturing System Integrator** *May 2025 – August 2025*
  *COOP* Marietta GA
  - Built an HMI display generator that produced 40+ P&ID screens for plant operators.
  - Reviewed and modified PLC programs across two production shops plus the PLC trainer system.
- **United States Marine Corps Reserve** *August 2022 - Present*
  *Ground Radio Repair Specialist* Marietta GA
  - Troubleshoot, maintain, and operate USMC radio systems in time-sensitive conditions; active Secret clearance.
  - Squad Leader of 8 Marines in the 4th Combat Readiness Regiment, responsible for training, accountability, and readiness.